

## Intro to Cloud Computing (CS4037) BCS9A, BDS7A

Date: November 12<sup>th</sup>, 2024

Course Instructor(s): Ali Khawaja

## Sessional-II Exam

Total Time (Hrs.): 1

Total Marks: 100

Total Questions: 11

Roll No

Section

Student Signature

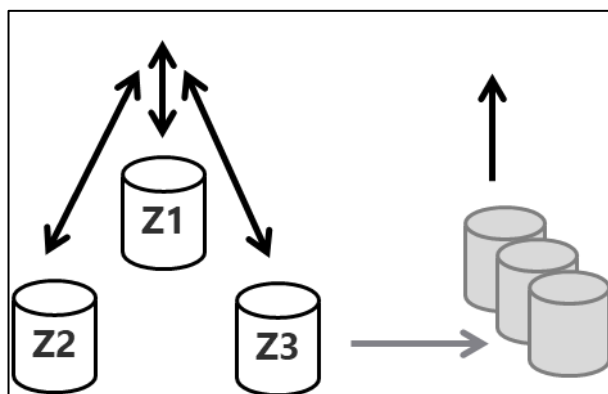
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Attempt all the questions. Provide concise and brief answers. Please answer only what is being asked.

### *CLO 1. Describe the basics of cloud computing and related services*

Q1: (a) Draw a figure with brief explanation to show storage account configured as RA-GZRS. (b) Why storage accounts need to have globally unique names? (10)

(a)



Geographically zone-redundant storage (GZRS) combines the high availability of ZRS with the protection from regional outages that GRS provides. A GZRS storage account replicates data across three Azure AZs in the primary region, and to a secondary geographic region for protection from regional disasters.

(b) Storage account name is part of the URL to access that storage account, and URLs must be globally unique for them to be accessible [since each URL has its own specific DNS entry]

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### ***CLO 1. Describe the basics of cloud computing and related services***

Q2: (a) Briefly describe the usage of blob access tiers. (b) What is the purpose of blob lifecycle management rules? (10)

| <br>Hot | <br>Cool | <br>Archive                          |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Optimized for storing data that is accessed frequently.                                  | Optimized for storing data that is infrequently accessed and stored for at least 30 days. | Optimized for storing data that is rarely accessed and stored for at least 180 days with flexible latency requirements. |

(b)

Transitioning of blobs to a cooler storage tier to optimize for performance and cost

Delete blobs at the end of their lifecycle

Apply rules to filtered paths in the Storage Account

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### ***CLO 1. Describe the basics of cloud computing and related services***

Q3: (a) Briefly describe the usage of azure blob and azure files. (b) Briefly provide the purpose of **azcopy** and **azure file sync**. (10)

| Feature            | Description                                                                                                                                      | When to use                                                                                                                                                                                                                                                   |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Azure Files</b> | SMB interface, client libraries, and a REST interface that allows access from anywhere to stored files                                           | <ul style="list-style-type: none"><li>• Lift and shift an application to the cloud</li><li>• Store shared data across multiple virtual machines</li><li>• Store development and debugging tools that need to be accessed from many virtual machines</li></ul> |
| <b>Azure Blobs</b> | Client libraries and a REST interface that allows unstructured data (flat namespace) to be stored and accessed at a massive scale in block blobs | <ul style="list-style-type: none"><li>• Support streaming and random-access scenarios</li><li>• Access application data from anywhere</li></ul>                                                                                                               |

| AzCopy                                              | Azure File Sync                                                             |
|-----------------------------------------------------|-----------------------------------------------------------------------------|
| Command line utility                                | Synchronizes Azure and on premises files in a bidirectional manner          |
| Copy blobs or files to or from your storage account | Cloud tiering keeps frequently accessed files local, while freeing up space |
| One-direction synchronization                       | Rapid reprovisioning of failed local server (install and resync)            |

### ***CLO 2. Apply cloud computing knowledge in various real world use cases (situations)***

Q4: (a) What is the **relationship** between Azure App Service and App Service Plan (ASP)? (b) What is the minimal ASP which allows you to have deployment slots and autoscaling? (c) When copying setting between deployment slots, how do you ensure a specific setting is not copied to the other slot (10 Points)

- (a) App Service Plan provides the **compute** and **features** for Azure app service.
- (b) Standard
- (c) You select an option (slot setting) to mark that setting as slot specific setting so it does not get copied.

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### ***CLO 2. Apply cloud computing knowledge in various real world use cases (situations)***

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Q5: (a) When creating a custom domain for the web app, when would you create (i) CNAME record? (ii) A record? (b) Briefly explain the difference between isolated and dedicated compute. (10)

***Note: this question is not what is CNAME or A record is. Its specifically asking when you would create either one when dealing with web apps. We covered this point in the class as well as in the review session.***

- (a) Non-Isolated ASPs: CNAME, Isolated Plan: A record
- (b) Isolated Plan has dedicated network as well besides dedicated compute.

### ***CLO 2. Apply cloud computing knowledge in various real world use cases (situations)***

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Q6: Given the address space: 10.0.0.0/8, (a) Divide the above into /20 subnets and provide the ranges. (b) Divide the above into /16 subnets and provide the ranges.

***Note: We covered this in class, and I provided detailed solution to this problem (with different ranges of course) during review.***

Given: /8:  $2^{24}$ : 16,777,216

#### **(a) Dividing /8 into /20 subnets**

/20:  $2^{12}$ : 4096 (size of each subnet)

$16777216/4096$ : 4096 → So we'll have total of 4096 /20 address spaces:

Range of each subnet:  $4096/256$ : 16

10.0.0.0 → 10.0.15.255

10.0.16.0 → 10.0.31.255

10.0.32.0 → 10.0.47.255 (and so on)

#### **(b) Dividing /8 into /16 subnets**

/16:  $2^{16}$ : 65536 (size of each subnet)

$16777216/65536$ : 256 → So we'll have total of 256 /16 address spaces:

Range of each subnet:  $65536/256$ : 256

10.0.0.0 → 10.0.255.255

10.1.0.0 → 10.1.255.255

10.2.0.0 → 10.2.255.255 (and so on)

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### ***CLO 2. Apply cloud computing knowledge in various real world use cases (situations)***

Q7: (a) Explain persistent volume claim and persistent volume in AKS. (b) When would you attach managed disk vs azure files to containers. (10)

(a) Persistent Volume (PV) is a storage resource created and managed by the Kubernetes API that can exist beyond the lifetime of an individual pod.

A persistent volume claim (PVC) requests storage of a particular storage class, access mode, and size.

(b) Note: This question is asking usage of managed disk vs azure files when dealing with containers. Its NOT asking a generic question regarding managed disk vs azure files. Again, I covered this specifically during class and during review as well.

Containers with managed disk: For container local usage

Containers with Azure Files: Central location for all containers to save application data

### ***CLO 2. Apply cloud computing knowledge in various real world use cases (situations)***

Q8: When comparing containers and VMs, briefly explain the difference between (a) Isolation, (b) Operating System, (c) Fault Tolerance (10)

| Feature          | Containers                                                                                                                                            | Virtual Machines                                                                                                                                                                                                  |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Isolation        | Typically provides lightweight isolation from the host and other containers but doesn't provide as strong a security boundary as a virtual machine    | Provides complete isolation from the host operating system and other VMs. This is useful when a strong security boundary is critical, such as hosting apps from competing companies on the same server or cluster |
| Operating system | Runs the user mode portion of an operating system and can be tailored to contain just the needed services for your app, using fewer system resources. | Runs a complete operating system including the kernel, thus requiring more system resources (CPU, memory, and storage)                                                                                            |
| Fault tolerance  | If a cluster node fails, any containers running on it are rapidly recreated by the <b>orchestrator</b> on another cluster node                        | VMs can fail over to another server in a <b>cluster</b> , with the VM's operating system restarting on the new server                                                                                             |

### ***CLO 2. Apply cloud computing knowledge in various real world use cases (situations)***

Q9: (a) What is the purpose of **mountPath** attribute in AKS manifest file? (d) Briefly explain the purpose of **kubeproxy** and **kubelet** in an AKS node. (5)

- **mountPath**: folder path where mounted storage will be made available
- **Kubeproxy**: Maintains network rules on nodes to implement [Services](#). Used to expose pods on the network.
- **Kubelet**: Ensures that Pods (and the containers defined in them) are running

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### ***CLO 2. Apply cloud computing knowledge in various real world use cases (situations)***

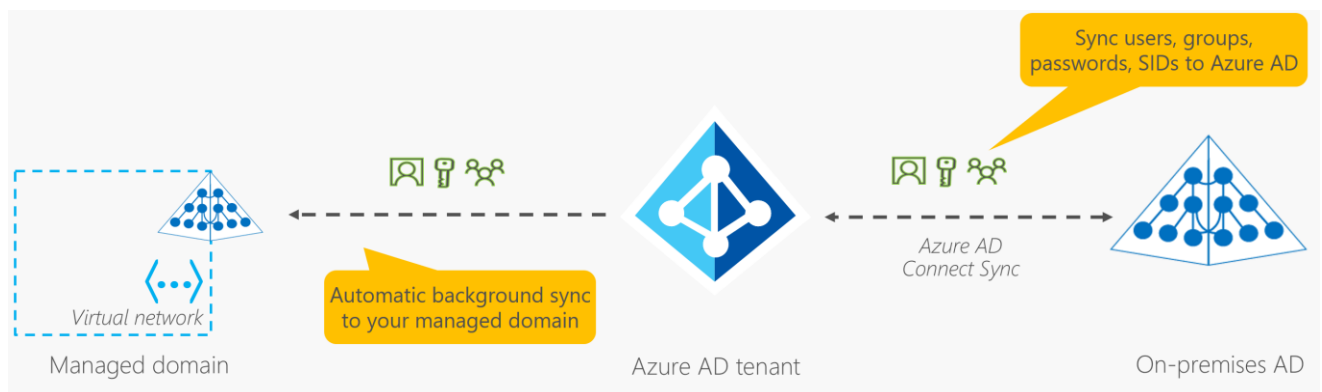
Q10: (a) What are the 3 options when synchronizing password hash between ADDS and Entra ID. (b) How and why secure score is calculated by defender for cloud? (5)

- (a) No sync, One way sync, 2 way sync
- (b) Score is calculated by giving each found vulnerability a rating/score, and it is done to provide a high level view of how secure the whole environment is on a 1-100 scale.

### ***CLO 2. Apply cloud computing knowledge in various real world use cases (situations)***

Q11: (a) What is the purpose of Entra ID and Entra Domain Services? (b) Draw & briefly explain the process of synchronization of ADDS objects from on-premises to Entra Domain Services in Azure. (10)

- (a) **Entra ID:** Http[s] based authentication and authorization service; **Entra Domain services:** PaaS Domain Controller to support legacy apps (which require LDAP based authentication)
- (b) OnPrem ADDS does bidirectional sync with Entra ID (Old Name: Azure AD), which then does one-way sync with Entra DS PaaS Service (Managed Domain)



### ***BONUS QUESTION***

### ***CLO 2. Apply cloud computing knowledge in various real world use cases (situations)***

Q12: (a) Briefly explain the concept of zero trust. (b) Even after fully authenticated and authorized, which service in Entra ID allows an administrator to block user access if he is not in a specific geographic location? (10)

- (a) Zero Trust is a new security model that assumes breach and verifies each request as though it originated from an uncontrolled network. It is based on the following principles: (i) Verify explicitly, (ii) Use least privilege access, (iii) Assume breach
- (b) Conditional Access